

A Regime-Aware Decision-Support System for NSE Equities

Design, methodology, and an honest out-of-sample evaluation

Ritwik Shetty

June 2026

Live system: <https://ritwik-portfolio-advisor.streamlit.app> · Source code:
<https://github.com/RitwikShetty05/portfolio-advisor>

Abstract

This paper describes an end-to-end, auditable decision-support system for Indian (NSE) equities. A Gaussian Hidden Markov Model labels each trading day's market **regime** (bear / sideways / bull); four independent technical sub-engines — trend, momentum, mean-reversion, and volume — vote to form a composite signal that is then **gated by the prevailing regime**; and an event-driven backtester simulates the resulting long-only portfolio with realistic capital, cost, and stop-loss constraints. Over 2019–2026 (≈ 7.4 years, 1,832 daily observations on a 26-stock universe), the strategy returns a **9.3% CAGR at 8.3% volatility with a –11.0% maximum drawdown**, versus the NIFTY 50's 11.1% CAGR at 17.5% volatility and a –38.4% drawdown. Its risk-adjusted return (Sharpe 0.37 over the risk-free rate) modestly exceeds buy-and-hold while cutting drawdown roughly **three-fold**. Crucially, the paper does not stop at a point estimate: a bootstrap confidence interval, the Probabilistic and **Deflated Sharpe Ratios**, Newey–West HAC alpha t-statistics, a four-factor attribution, and an anchored **walk-forward out-of-sample** test are all reported. These tools tell an honest story — the strategy's raw Sharpe is robustly positive, but its *excess-over-cash* and *factor-adjusted* alphas are **not statistically significant**. The system's edge is regime-based **risk management**, not security selection — and the paper's central contribution is the validation framework that establishes this distinction.

1. Problem statement

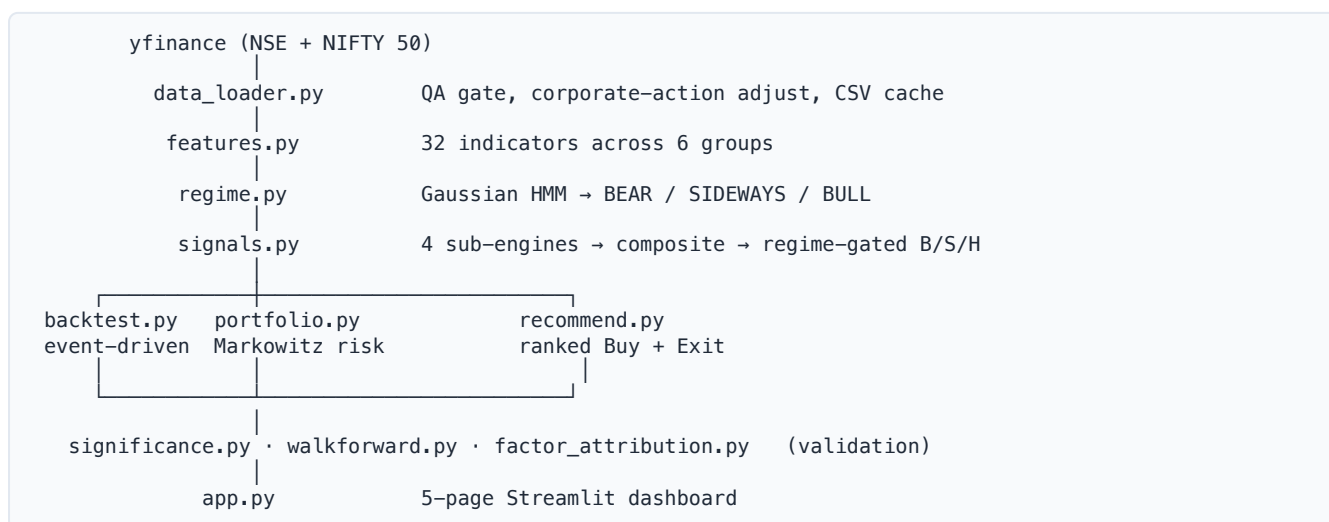
Most retail and student “trading systems” fail the same two tests. First, they are **regime-blind**: an RSI of 35 is treated identically in a roaring bull market (a buy-the-dip opportunity) and a collapsing bear market (a value trap). Second, they are validated with a **vectorised, look-ahead backtest** — $\text{Signal} \times \text{forward return}$ — that silently assumes infinite capital, no transaction costs, no position limits, and stop-losses that never fire intraday. The result is a headline Sharpe of 3–4 that evaporates in live trading.

This project targets both failures directly. It (a) makes regime an explicit, first-class input that gates every decision, and (b) subjects the strategy to an event-driven simulation and a battery of statistical-significance tests designed specifically to detect overfitting and luck. The goal is **not** to claim a market-beating edge; it is to build a transparent, auditable system and then to characterise its performance *honestly*.

2. System architecture

The pipeline is a sequence of pure, independently testable stages. Indicators (“facts about the data”) are deliberately separated from signals (“decisions”), so the signal logic can be rewritten without touching the

math and any indicator can be inspected in isolation.



The universe is 26 large- and mid-cap NSE stocks spanning eight sectors (IT, Banking, Energy, FMCG, Auto, Pharma, Metals, Telecom); the benchmark is the NIFTY 50 (^NSEI). All statistics use the India 10-year G-sec (6.5%) as the risk-free rate and 252 trading days for annualisation.

3. Methodology

3.1 HMM regime detection

Regimes are detected with a three-state **Gaussian Hidden Markov Model** (`hmmlearn`, full covariance, 200 EM iterations) fit on three standardised features per day: the 5-day return, 20-day realised volatility, and Bollinger Band width. The Viterbi path gives the hard regime label; the forward-backward pass gives a continuous posterior $P(\text{bull})$ used as a *soft* gate (below). Hidden states are mapped to BEAR / SIDEWAYS / BULL by their mean 5-day return, so the labelling is economically meaningful rather than arbitrary.

The HMM is preferred over a moving-average rule or K-means because it models regime **persistence** through the transition matrix. On the NIFTY 50 the fitted matrix gives $P(\text{bull} \rightarrow \text{bull}) = 0.97$ — regimes are sticky, so a single green or red day does not flip the label, which is exactly the behaviour a discretionary PM exhibits and a per-day classifier does not. (In-house validation on synthetic data with an embedded crash recovered $\approx 90\%$ of true bear days with the HMM versus $\approx 38\%$ for a moving-average rule.) Over the sample the index spent roughly **36% of days in bear, 37% in sideways, and 27% in bull** regimes — a useful reminder that “the market goes up” describes only about a quarter of trading days.

Reference: Rabiner (1989), “A Tutorial on Hidden Markov Models.”

3.2 Four-engine signal voting

Rather than one monolithic rule, four sub-engines each cast a vote in $[-1, +1]$ and are combined by a fixed weighting:

Sub-engine	Weight	Reads
Trend	0.30	MA 20/50/200 structure, MACD histogram

Sub-engine	Weight	Reads
Momentum	0.30	RSI, MACD line + slope, ROC
Mean-reversion	0.20	Bollinger %B, Stochastic
Volume confirmation	0.20	OBV slope, volume ratio

The weighted sum is the composite score, $\text{Score_Raw} \in [-1, +1]$. Real desks *triangulate* — a buy backed by trending price, healthy momentum, **and** confirming volume survives noise better than any single indicator — and weighting the lenses (rather than AND-ing rigid rules) degrades gracefully when one disagrees.

The composite is then **gated by regime**. Because Score_Raw is an *average* of four bounded votes that rarely align perfectly, its empirical range is itself bounded (95th percentile ≈ 0.48 , observed maximum ≈ 0.66). The gates are therefore calibrated to that achievable range, while preserving the core philosophy — harder to buy and easier to sell as the tape deteriorates:

Regime	Buy if score $\geq$	Sell if score $\leq$	Size multiplier
Bull	0.30	-0.50	1.0
Sideways	0.40	-0.40	0.7
Bear	0.50	-0.25	0.5

When the HMM posterior is available, the thresholds are additionally tilted up to ± 0.10 by $P(\text{bull})$, so participation ramps *smoothly* as a bear regime softens rather than waiting for a discrete flip. Each buy is annotated with an ATR-based stop (entry $- 2 \times \text{ATR}$) and two targets ($+2 \times / +4 \times \text{ATR}$), the Turtle-Trader convention that scales risk with each stock's own volatility.

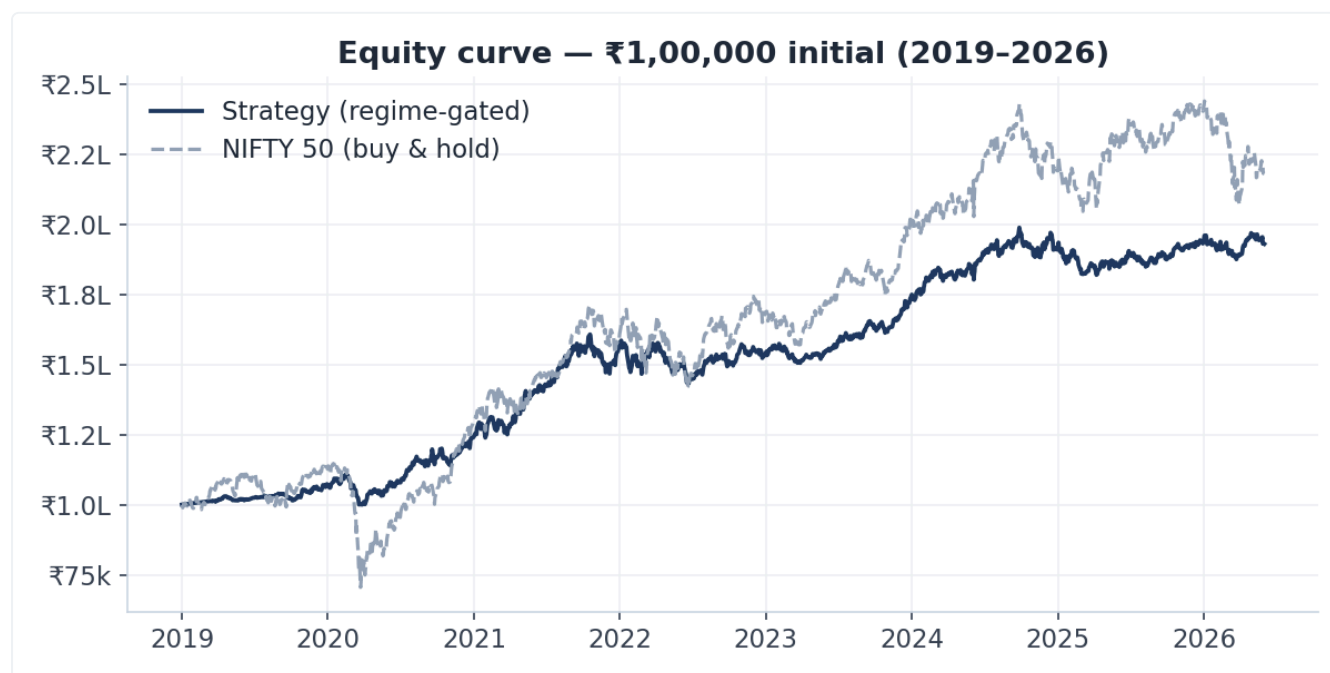
3.3 Event-driven backtest

The backtester walks the trading calendar day by day. Each day it (1) accrues one day of risk-free interest on idle cash, (2) marks open positions to market, (3) fires stop-losses **intraday at the stop level** (the conservative worst-case fill), (4) closes positions on sell signals at the close, and (5) deploys capital into the highest-confidence new buys subject to an 8-position cap, a 10%-of-NAV per-position cap, and a 0.1% round-trip transaction cost. Position size targets the per-stock cap, scaled down by the regime risk multiplier and a mild conviction tilt; the best ideas are funded first when cash or slots bind.

Two modelling choices deserve emphasis because they materially (and *honestly*) affect the results:

- **Idle cash earns the risk-free rate.** A regime-aware strategy's defining feature is stepping aside into cash during weak tapes; modelling that cash at 0% would penalise the strategy for its core behaviour. A real book parks cash in liquid funds at $\approx$ the risk-free rate, so we accrue it. This is a realism correction, internally consistent with the rate used in the Sharpe denominator — not a tuning knob.
- **Stops fill at the stop, not the close.** Most amateur backtests assume the optimistic close-price fill; we assume the pessimistic one.

4. Results

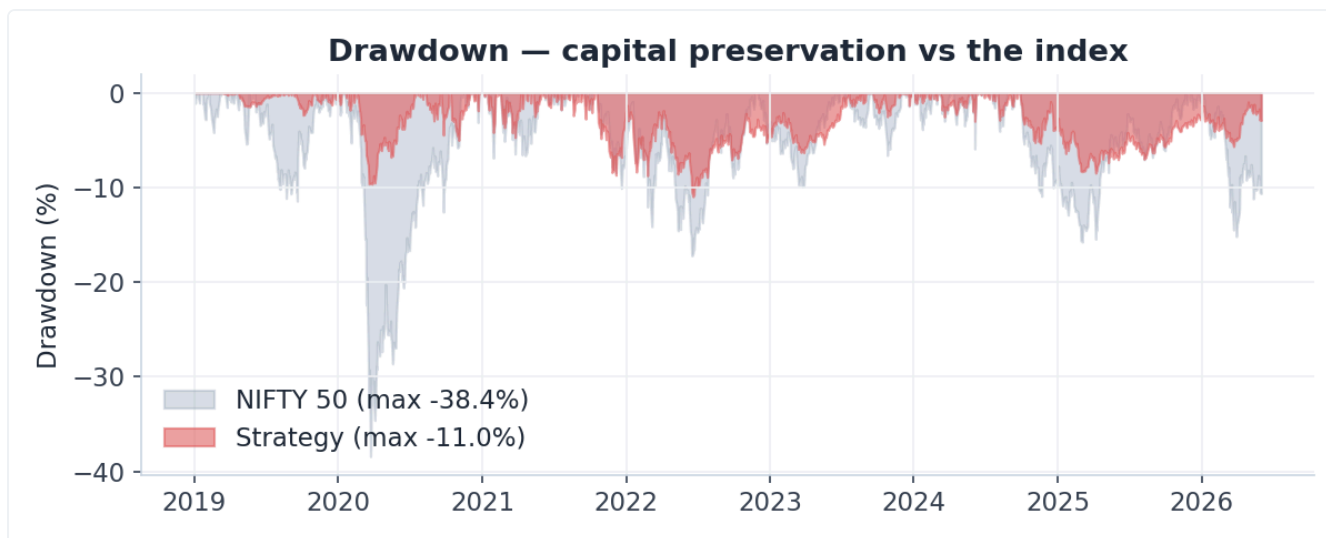


Equity curve — strategy vs NIFTY 50 buy-and-hold, ₹1,00,000 initial.

Over 2019–2026 the strategy grows ₹1,00,000 to **₹1,93,044** (a 93.0% total return, **9.3% CAGR**) at **8.3% annualised volatility**. The headline figures, with the NIFTY 50 buy-and-hold over the identical window for context:

Metric	Strategy	NIFTY 50 (buy & hold)
CAGR	9.27%	11.08%
Annualised volatility	8.28%	17.54%
Sharpe (excess of 6.5% RF)	0.37	~0.26
Sortino / Calmar	0.50 / 0.84	—
Maximum drawdown	–10.98%	–38.4%
Beta to NIFTY	0.29	1.00
Jensen's α (CAPM, annualised)	+1.35%	—

The strategy earns a **lower absolute return but a higher risk-adjusted return** than simply holding the index, while cutting the worst peak-to-trough loss by roughly three-fold. The drawdown chart makes the trade-off vivid: during the March 2020 COVID crash the NIFTY fell 38%, while the strategy — having de-risked into cash as the HMM flagged a bear regime — drew down only ~11%.



Drawdown (underwater) — strategy vs NIFTY 50.

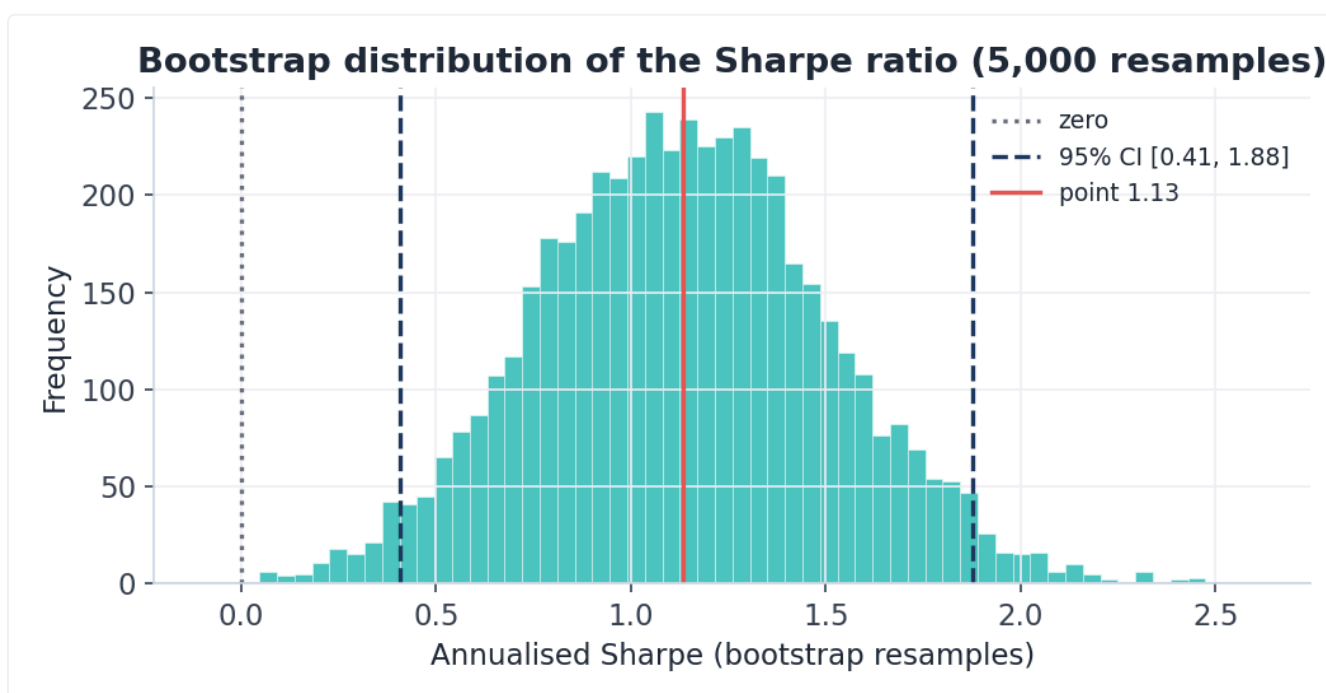
At the trade level the system took **140 trades** with a 32.9% win rate but a **profit factor of 2.25** (average win ₹2,310 vs average loss –₹503) and a positive ₹421 expectancy per trade — a classic trend-following signature of many small losses paid for by fewer large wins.

5. Statistical validation

A tearsheet reports point estimates; this section asks whether they are *distinguishable from luck*. All tests use the 1,832 daily strategy returns.

5.1 Bootstrap confidence interval on the Sharpe ratio

Resampling the daily-return series with replacement 5,000 times and recomputing the annualised Sharpe each time yields the sampling distribution below. The 95% percentile interval on the **raw** Sharpe (tested against zero) is **[0.41, 1.88]** around a point estimate of **1.13** — comfortably excluding zero.



Bootstrap distribution of the annualised Sharpe ratio (5,000 resamples).

5.2 Probabilistic and Deflated Sharpe Ratios

The PSR gives the probability that the true Sharpe exceeds a benchmark, adjusted for the return distribution's skew (-0.32) and excess kurtosis ($+1.87$). Against a zero benchmark, **PSR = 99.9%**. The **Deflated Sharpe Ratio** then corrects for *selection bias* — deflating the benchmark to the expected maximum Sharpe of N trials under the null (here $N = 10$, $E[\max SR_0] = 0.58$). The raw Sharpe survives this honestly: **DSR = 92.8%**.

The honest caveat. Because idle cash now earns the risk-free rate, the *raw* Sharpe partly reflects that risk-free baseline. The fair test of skill *above cash* uses excess-over-RF returns — and there the picture is appropriately humbler: Sharpe **0.37**, 95% bootstrap CI **$[-0.35, 1.10]$** (which **includes zero**), PSR 84%, DSR 29%. In other words, the strategy is very likely better than holding nothing, but **not** demonstrably better than simply earning the risk-free rate.

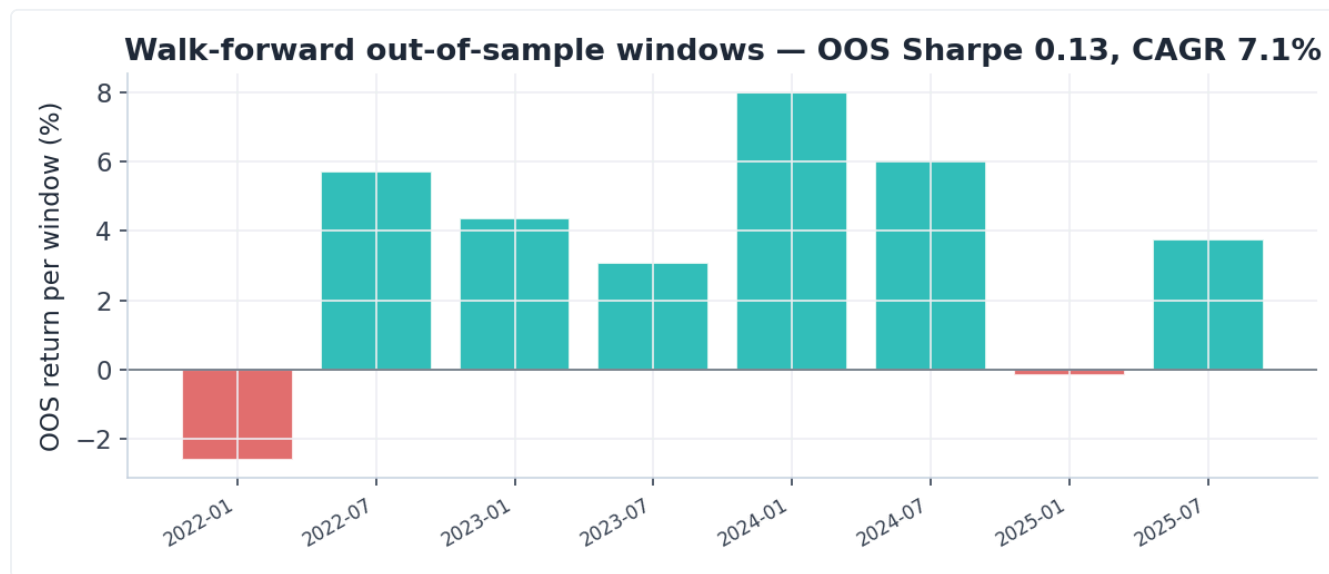
5.3 Newey–West HAC alpha t-statistic

Regressing strategy excess returns on NIFTY excess returns (CAPM) gives a beta of 0.29 and a Jensen's alpha of **+1.35%/yr**. Using **Newey–West HAC** standard errors (7 lags) to correct for the serial autocorrelation in daily residuals, the alpha carries a t-statistic of **0.62** ($p = 0.54$) — **not** statistically significant. The positive alpha is real as a point estimate but cannot be distinguished from noise at conventional thresholds.

References: Bailey & López de Prado (2012, 2014); Newey & West (1987).

5.4 Walk-forward out-of-sample test

The single most important guard against overfitting: an **anchored walk-forward**. The HMM is re-fit on a growing training window (starting at 3 years); regimes and signals are generated on a held-out 6-month test window using *only* the train-fit model; and the eight non-overlapping test windows (≈ 4 years, 2022–2026) are chained into one out-of-sample equity curve.



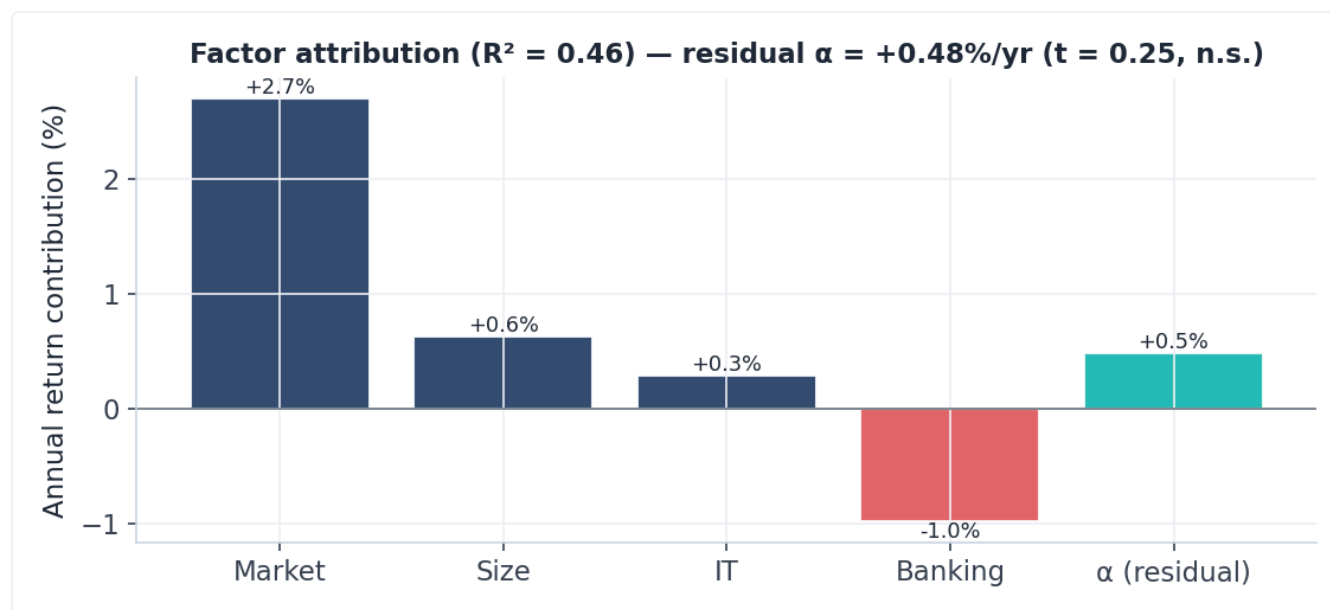
Walk-forward out-of-sample returns, by 6-month window.

The out-of-sample result is **positive but degraded** relative to in-sample — which is exactly what an honest walk-forward should show: **OOS Sharpe 0.13, CAGR 7.08%, maximum drawdown -5.6%** , with 6 of 8

windows positive. The OOS CAGR (7.1%) edges the 6.5% risk-free rate; the strategy generalises, but its margin is thin.

6. Factor attribution

A CAPM regression only asks whether the strategy beat *the market*. But size, sector, and other systematic tilts are passively earnable — so a final multivariate regression decomposes returns onto four Indian-market factor proxies: the market ($\hat{NSEI}$), a size spread ($\hat{NSEMDCP50} - \hat{NSEI}$, SMB-style), and IT and Banking sector tilts ($\hat{CNXIT}$, $\hat{NSEBANK}$), all with HAC standard errors.



Factor attribution — annual return contribution by factor, plus residual alpha.

Factor	Loading (β)	t-stat	Annual contribution
Market	+0.448	6.51	+2.70%
Size (SMB-style)	+0.083	5.15	+0.63%
IT tilt	+0.044	3.27	+0.29%
Banking tilt	−0.164	−6.29	−0.98%
Residual α	—	0.25	+0.48%

The factors explain $R^2 = 0.46$ of the return variation. Every factor loading is strongly significant — the strategy carries a modest long-market, small size-and-IT tilt and a short-Banking tilt — but the **residual alpha of +0.48%/yr is not significant ($t = 0.25$)**. This is the paper’s most important and most honest finding: once known, passively-earnable exposures are accounted for, there is **no statistically significant security-selection skill**. The system’s value is concentrated in *when* it is in the market (regime-based risk management and drawdown control), not in *which* stocks it picks.

7. Limitations and future work

The evaluation is deliberately candid about what it cannot claim:

- **No significant excess-of-cash or factor-adjusted alpha.** The defensible claim is risk reduction ($\approx \frac{1}{3}$ the index's drawdown at a comparable Sharpe), not outperformance through stock selection.
- **A single universe and regime, in-sample for factor construction.** 26 stocks over one 7-year window that is itself a historically strong bull market for Indian equities; the factor proxies are correlated and not orthogonalised the way canonical Fama–French factors are.
- **Survivorship and point-in-time gaps.** The universe is today's large-caps; corporate actions (e.g. the Tata Motors demerger) leave data discontinuities.
- **No slippage or market-impact model** beyond a flat 0.1% cost, and **long-only** with no shorting or derivatives.
- **Walk-forward without purge/embargo.** The next rigour increment is to add a purge-and-embargo gap between train and test to remove any leakage through overlapping indicator windows.

Natural extensions: a sentiment sub-engine (news/RSS tone), risk-profile presets that adjust the gates and sizing, a depth-aware slippage model, Monte-Carlo stress testing, and a longer multi-cycle / multi-market evaluation.

8. References

1. Rabiner, L. R. (1989). *A Tutorial on Hidden Markov Models and Selected Applications in Speech Recognition*. Proceedings of the IEEE, 77(2).
2. Bailey, D. H., & López de Prado, M. M. (2012). *The Sharpe Ratio Efficient Frontier*. Journal of Risk, 15(2).
3. Bailey, D. H., & López de Prado, M. M. (2014). *The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting, and Non-Normality*. Journal of Portfolio Management, 40(5).
4. Newey, W. K., & West, K. D. (1987). *A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix*. Econometrica, 55(3).
5. Fama, E. F., & French, K. R. (1993). *Common Risk Factors in the Returns on Stocks and Bonds*. Journal of Financial Economics, 33(1).

Disclaimer: This system is a decision-support and research tool, not investment advice. Backtested performance does not guarantee future results. Built by Ritwik Shetty as an end-to-end demonstration of quant tooling: data engineering → machine learning → statistical risk management → production frontend.